Real-Time Two-Way Sign Language Translation System

Bridging the Communication Gap Using AI & Computer Vision

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Project Guide

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The Communication Barrier

Over 5% of the global population (430 Million) suffers from disabling hearing loss, facing isolation in everyday public interactions.

Defining the Problem

The Daily Struggle

Deaf individuals face massive challenges in critical sectors like Banks, Hospitals, and Police Stations where officials lack Sign Language knowledge.

This leads to a loss of independence and privacy.

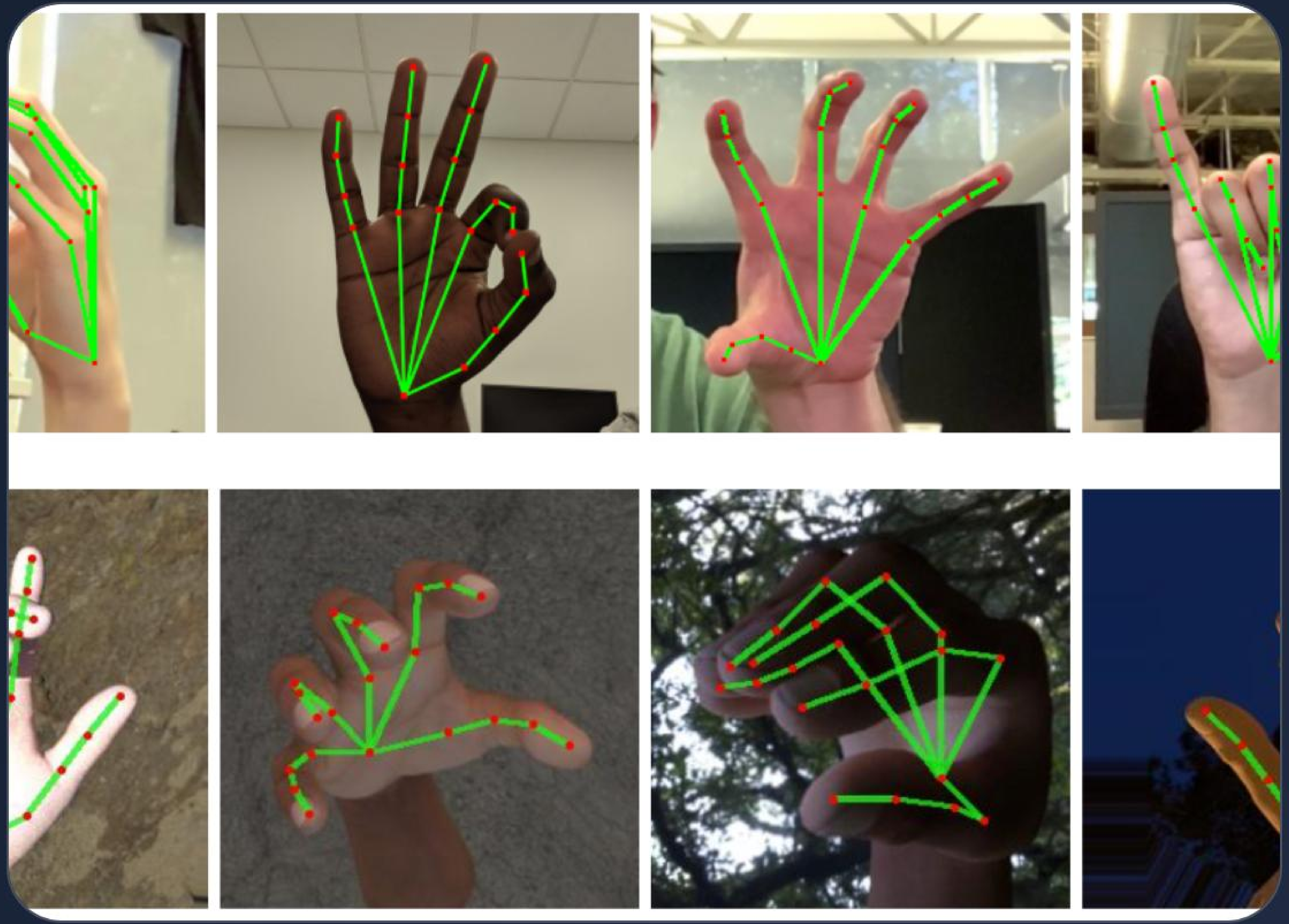
Current Limitations

- Human interpreters are expensive.
- Professional services are scarce.
- Interpreters are not available 24/7.
- High technical friction for apps.

Our Proposed Solution

AI-Powered Real-Time Translator

- **Hardware-Free:** Operates using a standard laptop webcam, no "smart gloves" required.
- **Two-Way Flow:** Converts Hand Gestures to Audio and Speech Input to Text display.
- **Accessibility:** Built specifically for public reception desks and service counters.



Project Objectives

Real-Time Detection

Process gestures with less than 1-second latency for natural flow.

High Accuracy

Achieve >90% accuracy using Deep Learning for dynamic gestures.

Zero Hardware Cost

Runs on existing office PCs without needing GPU farms.

Social Inclusion

Aligning with the Rights of Persons with Disabilities Act (2016).

Technology Stack

Category	Technology Selection	Role in System
Programming	Python 3.9+	Core logic and model integration
Computer Vision	MediaPipe & OpenCV	Skeletal hand tracking & video capture
Deep Learning	LSTM Neural Networks	Temporal gesture sequence prediction
Audio Output	pyttsx3 Library	Real-time Text-to-Speech conversion

System Methodology

Input

Standard Webcam captures 30 FPS video frames.

Extraction

MediaPipe extracts 21 key landmarks (x, y, z).

Analysis

LSTM model analyzes 30-frame motion patterns.

Output

Result displayed as text and spoken via speakers.

Why LSTM? (Temporal Intelligence)

>90%

Target Accuracy

Sequence vs Snapshots

Unlike standard CNNs that look at static photos, **LSTM (Long Short-Term Memory)** is designed for time-series data.

Sign language is movement-based. A wave implies "Hello"—it's a sequence. LSTM remembers past frames to understand the current intent.

Smartphone app for sign language translation

Scope & Future

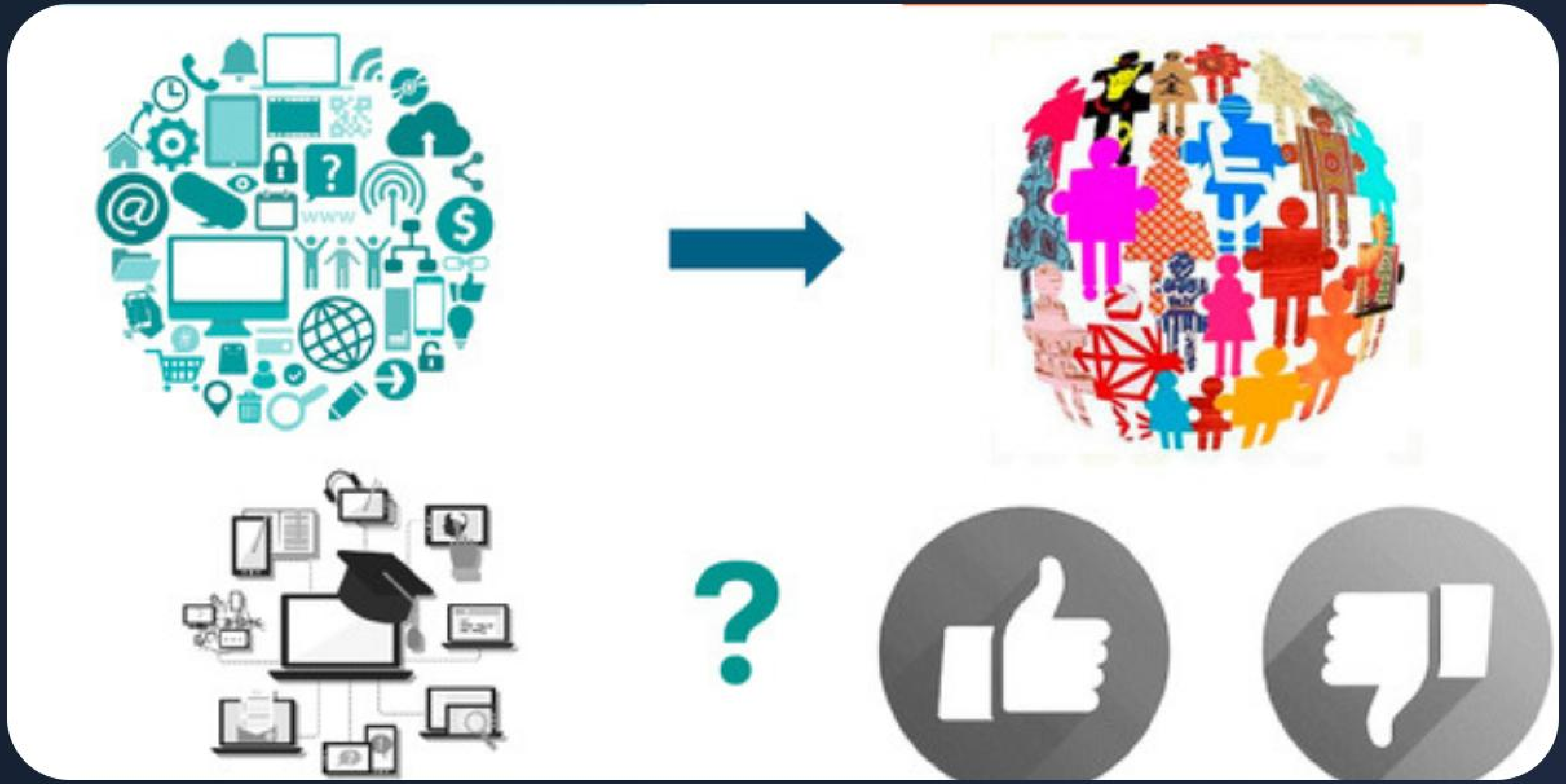
Expanding Accessibility

Our initial prototype focuses on banking and emergency vocabulary like "Name," "Withdraw," and "Help."

- ✓ Mobile App: Porting via TensorFlow Lite.
- ✓ Regional Languages: Support for Hindi/Punjabi.
- ✓ 3D Avatars: Digital characters that sign back.

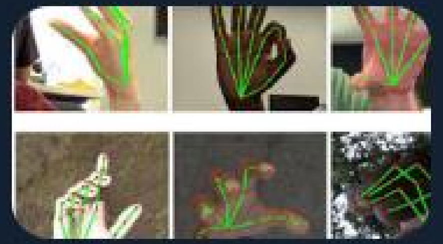
Thank You

Towards an Inclusive Digital India



Any Questions?

Image Sources



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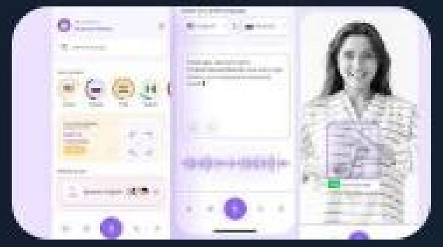
Source: research.google



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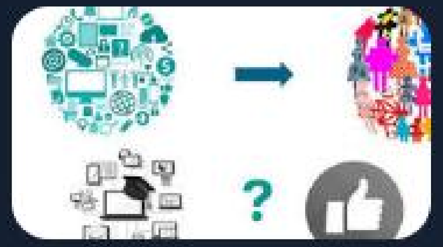

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